

## **Basic Information**

**Experiment No.:** 14

**Date:** 24-09-2026

**Title:** ListIterator Methods in Java

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## **Objective / Aim**

To demonstrate the "ListIterator" methods such as "next()", "set()", "add()", "previous()", and "remove()" using an "ArrayList".

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## **System Requirements**

### Hardware

- Computer/Laptop
- Minimum 4 GB RAM

### Software

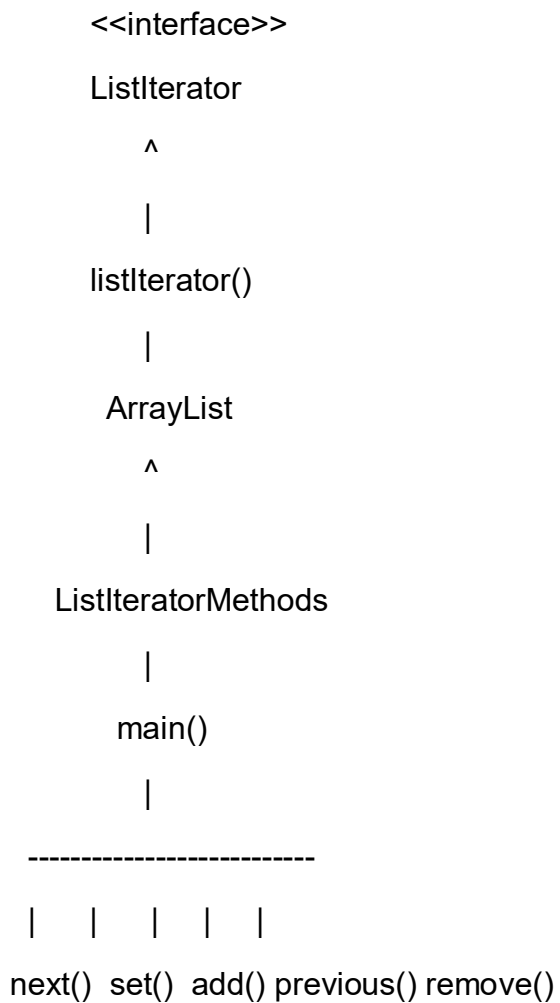
- Java JDK
  - Any Java IDE or Command Prompt
  - Operating System: Windows/Linux/macOS
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## **Algorithm / Logic Description**

1. Create an "ArrayList" and add three elements.
2. Create a "ListIterator" using "listIterator()".
3. Move the iterator to the required position using "next()".
4. Replace an element using "set()".
5. Insert a new element using "add()".

6. Move backward using "previous()".
  7. Remove an element using "remove()".
  8. Display the list after each operation.
  9. End the program.
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## UML Class Diagram



## Source Code

```
package collectionframework;
```

```
import java.util.List;
import java.util.ArrayList;
import java.util.ListIterator;

public class ListIteratorMethods {
    public static void main(String[] args) {

        List<String> list = new ArrayList<>();

        list.add("Apple");
        list.add("Banana");
        list.add("Mango");

        ListIterator<String> it = list.listIterator();

        // Move to Banana
        it.next();
        it.next();

        // set()
        it.set("Orange");

        System.out.println("After set: " + list);

        // add()
        it.add("Grapes");

        System.out.println("After add: " + list);
```

```
// Move to Grapes
it.previous();

// remove()
it.remove();

System.out.println("After remove: " + list);
}
}
```

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## **Compilation & Execution Commands**

Save the file as:

ListIteratorMethods.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

javac ListIteratorMethods.java

Run:

java collectionframework.ListIteratorMethods

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## **Output Screen**

After set: [Apple, Orange, Mango]

After add: [Apple, Orange, Grapes, Mango]

After remove: [Apple, Orange, Mango]

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## **Result**

The "ListIterator" methods "next()", "set()", "add()", "previous()", and "remove()" were successfully demonstrated using an "ArrayList".